

LAB 10: TESTING & DEPLOYMENT (CLOUD)

CloudDeploy – CI/CD Pipeline

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Project: Sendit-api

GitHub Repository: <https://github.com/winnie-Jemutai/sendit-api>

Render link: <https://sendit-api-i0qd.onrender.com/>

<https://dashboard.render.com/>

EXERCISE 1: INTEGRATION TESTS

Question 1: How is an integration test different from a unit test?

A unit test tests a small individual part of an application, such as a single function or component. An integration test checks whether multiple parts of the application work correctly together. For example, an integration test can test the complete process of registering a user, logging in, creating a document, updating it, and deleting it.

Question 2: Why is it important to test the full flow?

Testing the full flow is important because it verifies that all the connected parts of the API work correctly together. It can identify problems that may not be found when individual functions are tested separately and helps ensure that users can successfully complete the intended operations.

EXERCISE 2: LOAD TESTING

Question 3: How many requests per second can your API handle?

The number of requests per second depends on the results of the performance or benchmark test. The actual value should be taken from the load test results rather than estimated.

Question 4: What is the bottleneck?

A bottleneck is the part of the system that limits the API's performance. Possible bottlenecks include database queries, network requests, CPU usage, memory usage, or slow external services.

Question 5: How would you improve performance?

Performance can be improved by optimizing database queries, adding database indexes, using caching, reducing unnecessary network requests, improving the application code, and using appropriate server resources.

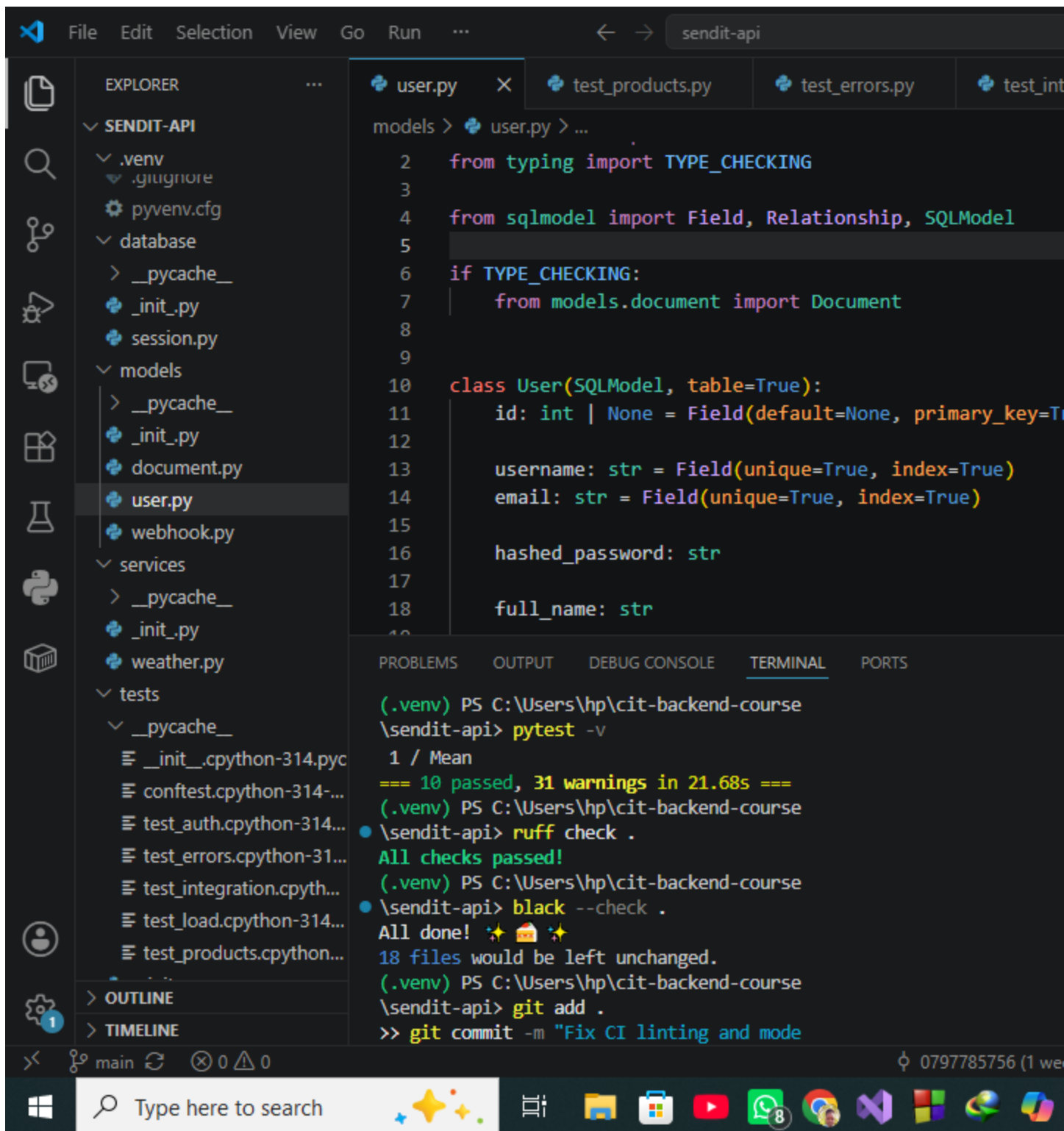
EXERCISE 3: CI/CD PIPELINE**Question 6: Why should you run tests on multiple Python versions?**

Testing on multiple Python versions helps ensure that the API works correctly across different supported Python environments. It can identify compatibility problems that might occur in one Python version but not another.

Question 7: Why is it important to run linters in CI?

Linters automatically check the code for errors, formatting problems, unused imports, and other code-quality issues. Running them in CI helps maintain consistent, clean, and reliable code before it is deployed.

REQUIRED SCREENSHOTS**Screenshot 1 – Pytest , Ruff, Black**



Screenshot 2 – GitHub Actions

Workflow run deleted successfully.

Actions

All workflows

CI/CD Pipeline

Management

- Caches
- Attestations
- Runners
- Usage metrics
- Performance metrics

CI/CD Pipeline

ci.yml

5 workflow runs

- ✓ **Fix CI linting and model relationships**
CI/CD Pipeline #14: Commit [68fb140](#) pushed by [winnie-Jemu](#)
- ✓ **Add load testing benchmark**
CI/CD Pipeline #8: Commit [2f306d3](#) pushed by [winnie-Jemu](#)
- ✓ **Add health and metrics monitoring**
CI/CD Pipeline #7: Commit [c0aa95c](#) pushed by [winnie-Jemu](#)
- ✓ **Complete Docker and CI testing setup**
CI/CD Pipeline #6: Commit [06a73f4](#) pushed by [winnie-Jemu](#)
- ✓ **Fix test database and passing tests**
CI/CD Pipeline #5: Commit [c8d8c1d](#) pushed by [winnie-Jemu](#)

Windows taskbar with search bar and application icons.

Screenshot 3– GitHub Repository

SendIt API - Swagger UI | sendit-db · Database · Render | winnie-Jemutai/sendit-api

github.com/winnie-Jemutai/sendit-api

winnie-Jemutai / sendit-api

Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings

 **sendit-api** Public

main 1 Branch 0 Tags Go to file

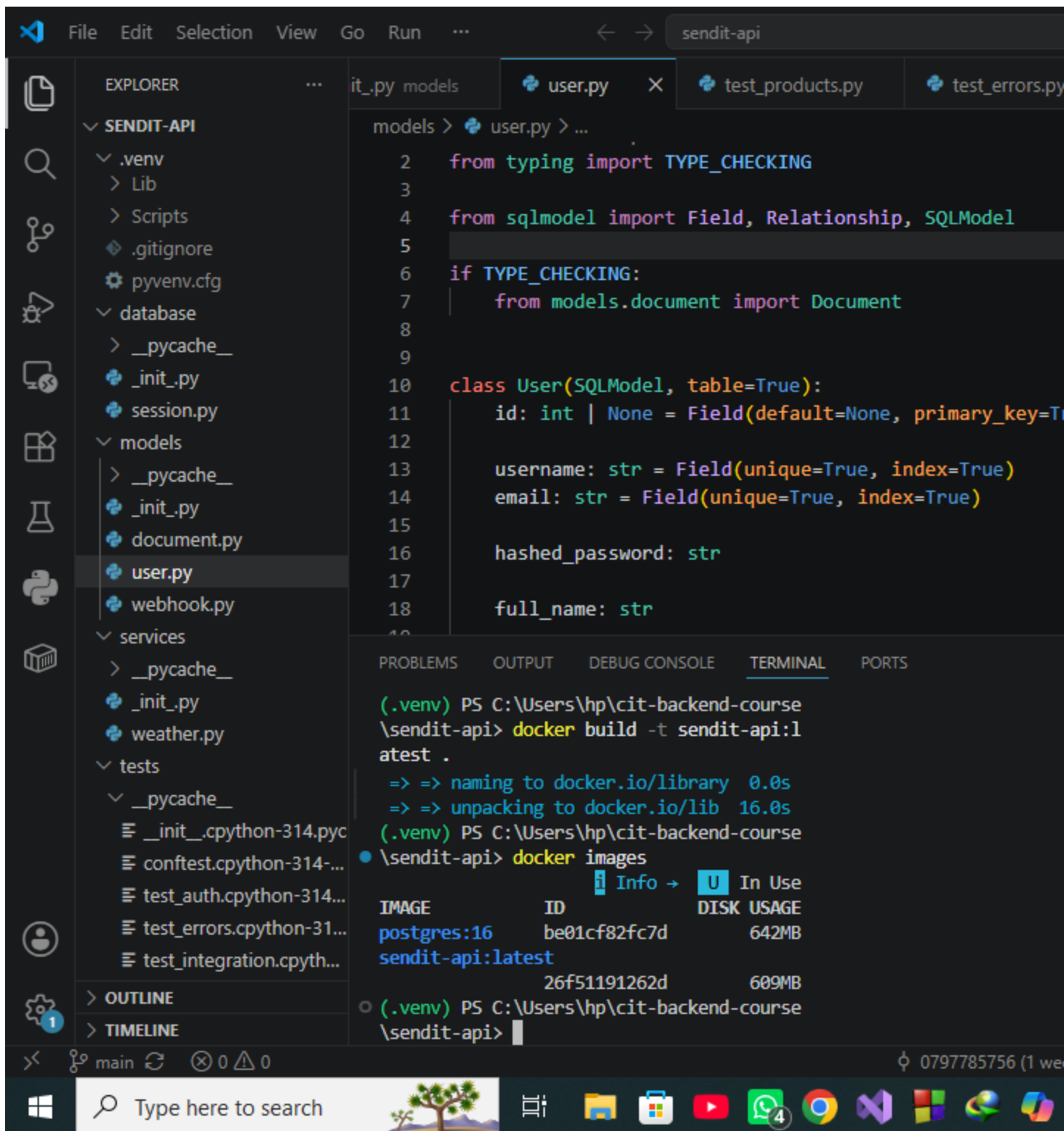
 **winnie-Jemutai** Fix CI linting and model relationships ✓

 .github/workflows	Fix GitHub Actions workflow
 __pycache__	Fix linting and formatting
 database	Fix linting and formatting
 models	Fix CI linting and model relationships
 services	Fix linting and formatting
 tests	Fix linting and formatting
 uploads	Fix CI linting and model relationships
 .env	Complete SendIt API Lab
 Dockerfile	Complete Docker and CI testing setup

Windows search bar: Type here to search

Taskbar icons: File Explorer, Calendar, YouTube, WhatsApp (4 notifications), Chrome, VS Code, Edge, and other applications.

Screenshot 4 – Docker



Screenshot 5– Render

CI Pipeline · Workflow runs · win x sendit-api · Web Service · Rend x SendIt API - Swagger UI

dashboard.render.com/web/srv-d9rn6im7bikc738pa4pg/deployments/dep-d9rn6iu7bikc738pa53

My Workspace sendit-api

Dashboard

sendit-api

Events

Settings

MONITOR

Logs

Metrics

MANAGE

Environment

Shell

Changelog

Invite a friend

Contact support

Render Status

August 8, 2026 at 9:19 PM ✓ Live

[06a73f4](#) Complete Docker and CI testing setup

All logs Search logs

Aug 8

09:21:11 PM	[4bn5c]	2026-08-08 18:21:11,559	INFO	
09:21:11 PM	[4bn5c]	INFO:		Application startup
09:21:11 PM	[4bn5c]	INFO:		Uvicorn running on
09:21:12 PM	[4bn5c]	INFO:		127.0.0.1:36852 - "
09:21:21 PM	[stqlw]	INFO:		Shutting down
09:21:21 PM	[stqlw]	INFO:		Waiting for applica
09:21:21 PM	[stqlw]	INFO:		Application shutdown
09:21:21 PM	[stqlw]	INFO:		Finished server pro
09:21:22 PM				==> Your service is live 🎉
09:21:22 PM				==>
09:21:22 PM				==> //////////////////////////////////////
09:21:22 PM				==>
09:21:22 PM				==> Available at your primary
09:21:22 PM				==>
09:21:22 PM				==> //////////////////////////////////////
09:21:22 PM	[4bn5c]	INFO:		10.28.128.130:0 - "

Need better ways to work with logs? Try the

Type here to search

Taskbar icons: Windows, Search, Task View, File Explorer, Calendar, YouTube, WhatsApp, Chrome, VS Code, Edge, and other applications.

Screenshot 6– Health Endpoint

CI Pipeline · Workflow runs · win x | sendit-api · Web Service · Rend x | SendIt API - Swagger UI

127.0.0.1:8000/docs#/default/health_check_health_get

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/health' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/health
```

Server response

Code	Details
200	Response body <pre>{ "status": "healthy", "timestamp": "2026-08-08T18:57:44.287520", "version": "1.0.0" }</pre> Response headers <pre>content-length: 79 content-type: application/json date: Sat,08 Aug 2026 18:57:43 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response

Windows taskbar: Type here to search, File Explorer, Calendar, YouTube, WhatsApp (8), Chrome, VS Code, Edge, etc.

Screenshot 7 – Render Logs

SendIt API - Swagger UI sendit-db · Database · Render

dashboard.render.com/d/dpg-d9rmojafngtc73dokd4g-a/logs?r=1h

My Workspace My project Production sendit-db

Environment

sendit-db

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Search logs

Aug 10

02:42:51 PM [6a79b930.20e9d-4] user=postgres,db=sendit-db time: 0:00:02.198 user=postgres database=sendit_db_fqdg SS

02:43:44 PM [6a79b970.26eda-1] user=[unknown],db=[unknown] host=10.27.148.170 port=46578

02:43:44 PM [6a79b970.26edb-1] user=[unknown],db=[unknown] host=10.27.148.170 port=46582

02:43:44 PM [6a79b970.26edc-1] user=[unknown],db=[unknown] host=10.27.148.170 port=46596

02:43:45 PM [6a79b970.26edc-2] user=postgres,db=sendit-db identity="postgres" method=md5 (/pgdata/

02:43:45 PM [6a79b970.26edc-3] user=postgres,db=sendit-db user=postgres database=sendit_db_fqdg SS

02:43:48 PM [6a79b970.26edc-4] user=postgres,db=sendit-db time: 0:00:03.091 user=postgres database=sendit_db_fqdg SS

02:43:48 PM [6a79b974.26ede-1] user=[unknown],db=[unknown] host=10.27.148.170 port=42336

02:43:48 PM [6a79b974.26edd-1] user=[unknown],db=[unknown] host=10.27.148.170 port=42326

02:43:49 PM [6a79b975.26edf-1] user=[unknown],db=[unknown] host=10.27.148.170 port=42342

02:43:49 PM [6a79b975.26edf-2] user=postgres,db=sendit-db identity="postgres" method=md5 (/pgdata/

02:43:49 PM [6a79b975.26edf-3] user=postgres,db=sendit-db user=postgres database=sendit_db_fqdg SS

02:43:51 PM [6a79b975.26edf-4] user=postgres,db=sendit-db time: 0:00:02.436 user=postgres database=sendit_db_fqdg SS

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Conclusion:

The SendIt API was tested using automated pytest tests, checked using Ruff and Black, integrated with GitHub Actions for CI/CD, and deployed to the cloud using Render.